PM approval by OTP: batch reopen and Unlock VLB

Final go-live note | Caliber Pulse-II (pulse-ll.cmll.in) | 02 November 2026 | Addendum to the two-stage VLB + ELB batch

1. What this adds

Two batch controls needed the site Project Manager to sign in to Pulse-II: approving a request to reopen a closed batch, and unlocking VLB after the tipper side was locked. From this release both can be approved by phone. The MIS user requests a one-time password, the PM receives it on Telegram (or WhatsApp), reads it out, and the MIS user enters it. The server records that as the PM's approval and acts at once. On-screen approval is unchanged and stays available.

One rule changed on the owner's instruction: a user who is both the site PM and the MIS requester may now approve their own reopening request on screen. The audit shows the same login as requester and approver.

> **Scope**
> This note covers only the approval OTP and the self-approval change. The two-stage batch itself, the express transfer OTP and the new-equipment OTP are already live (release of 01 November 2026) and are unchanged.

2. The flows

Reopen a closed batch by OTP

1. MIS user opens the closed batch, presses Request PM approval, types the reason (5 to 2000 characters).
2. Presses Request reopen OTP. The server creates the reopening request (or reuses the pending one), then sends the code to every current Project Manager of the batch site: Telegram first, WhatsApp to the registered mobile if Telegram is not linked.
3. MIS user phones the PM; the PM reads the six-digit code. MIS user enters it and presses Verify OTP.
4. The reopening request is marked approved (decided by the first PM the code went to, note "Approved by OTP sent to …"), and the batch is reopened immediately for that MIS user, at the machine stage, exactly as the Reopen button does.
5. If the reopen is blocked for another reason (another open batch at the site), the approval is kept and the Reopen approved batch button works once the blocker is cleared.

Unlock VLB by OTP

1. At the machine stage the batch owner or any MIS user presses Unlock VLB (PM OTP), next to the PM-only Unlock VLB.
2. Types the reason, presses Request unlock OTP, phones the PM, enters the code, presses Verify OTP.
3. VLB unlocks: stage returns to 1 VLB, the machine snapshot is cleared and rebuilt at the next Lock VLB. Machine entries already saved stay in the Equipment Log Book.

Unchanged

- PM approves or rejects on screen in Pulse-II (reopen requests list or the batch). Rejection still needs a remark.
- PM or MIS administrator presses Unlock VLB directly, with a reason.

- During the owner-authorized PM trial no approval is needed, so the OTP buttons are hidden.

3. Controls

Control	Value
Code	Six digits, random. Only a SHA-256 hash of id + code is stored.
Validity	10 minutes.
Wrong attempts	5, then the code is cancelled and a new one must be requested.
Resend	One request per minute per batch, purpose and requester. A new request cancels the previous live code.
Who can enter it	Only the MIS user who requested it, and only against the batch it was issued for. Another user gets "no longer available".
Who receives it	Every current Project Manager of the batch site, as resolved from the Users master (same list as the Vehicle transfers workflow).
Delivery	Telegram through the MIS bot (linked chats), WhatsApp fallback through the approved OTP template. No channel: the request fails with a clear message and the on-screen path remains.
Single use	A used code answers "already used".
Reload	The requester's live code is returned with the batch, so closing the dialog or reloading the page does not lose it.

4. Audit trail

Audit action	Written when	Detail
approval_otp_sent	A code is issued	purpose (reopen or unlock_vlb), reopen request id, PM logins, channels, reason
reopen_approved	Reopen OTP verified	via: otp, request id, requester, PM logins, note
reopened	Batch reopened	approvedBy = PM login recorded on the request (unchanged)
vlb_unlocked	Unlock OTP verified (or on-screen unlock)	via: otp or screen, approvedBy = PM logins (OTP only), reason

The reopening request row itself carries decided_by (the first PM the code went to) and decision_note "Approved by OTP sent to pm1, pm2", so the existing approvals screen shows how it was approved.

5. Technical summary

Server

- vlb-approval-otp.mjs (new): schema, message text, issue / consume / live-code helpers.
- vlb-batches.mjs: three routes, reopen request and reopen bodies factored into ensureReopenRequest and performReopen, self-approval check removed, approvalOtps and canRequestUnlockOtp in the batch detail.
- vlb-batch-stages.mjs: unlockVlb records via and approvedBy in the audit.
- vlb-batch-policy.mjs: appends the new schema (table mis_vlb_approval_otps). log-book.mjs and server.mjs: pass the PM lookup and OTP delivery already used by the express transfer.

```
POST /api/vlb-batches/:id/approval-otp              {purpose: reopen | unlock_vlb, reason}
POST /api/vlb-batches/:id/approval-otp/:otpId/verify {otp}
GET  /api/vlb-batches/:id/approval-otp              -> live codes of the caller
Table mis_vlb_approval_otps: id, batch_id, purpose, request_id, closure_revision, reason,
  requested_by, sent_to[], channels[], otp_hash, expires_at, attempts, status, used_at,
approved_by
```

Screens

- src/vlb-approval-otp.jsx (new): ApprovalOtpPanel (request, countdown, code entry, resend) and VlbUnlockOtpDialog.
- src/vlb-reopen-dialog.jsx: Request reopen OTP next to Request PM approval, and again while the request is pending.
- src/vlb-create-batch.jsx: Unlock VLB (PM OTP) button at the machine stage for non-PM users.

Permissions

All routes sit under /api/vlb-batches and inherit the Vehicle Log Book permission, like the other batch routes. No new permission or menu item.

6. Tests

Run on this machine on 02 November 2026 before the commit.

Test file	What it proves	Result
test/vlb-approval-otp.test.mjs (new)	Message text; unlock OTP: nothing to unlock before lock, reason required, no delivery channel, one per minute, only the requester enters it, wrong code counts down, success unlocks and audits via otp, reuse refused. Reopen OTP: PMs cannot request, same request as the screen, success approves and reopens at the machine stage with decided_by and note, approval kept when another open batch blocks the reopen, Reopen button works after. Dual-role self-approval on screen.	2 of 2 pass
test/vlb-approval-otp-ui.test.mjs (new)	Panel shows the request first, then code entry with countdown, resend and the right aria label; unlock dialog hides the second reason box while a code is live; reopen dialog offers the OTP next to the on-screen request, again while pending, never during the PM trial.	2 of 2 pass
test/vlb-batch-governance.test.mjs (changed)	The former 'no self approval' assertion now expects 200 and decided_by = the dual-role login; every other governance rule unchanged.	3 of 3 pass
test/vlb-batches.test.mjs, vlb-batches-ui.test.mjs, vlb-batch-concurrency.test.mjs, vlb-two-stage.test.mjs, vlb-two-stage-ui.test.mjs, express-otp.test.mjs	Existing batch, reopen, two-stage and express OTP behaviour unchanged.	all pass
Full suite (npm test)	3197 tests: 3174 pass, 4 fail. The 4 are the known Windows-only environment failures (python3 missing for the Gemini check, symlink permission in the release package, a busy temp folder in the startup gate, and a CRLF regex in one deletion UI test). None touches this change; all pass on the Linux runners.	3174 / 3197
npx vite build	Front end builds with the new dialog and panel.	ok

Everything specific to this change passes. The four full-suite failures are the same environment-only failures seen before the 01 November release and pass in GitHub Actions.

7. Go-live steps

1. Commit and push to main. GitHub Actions runs the test suite and deploys to pulse-ll.cmll.in; every user is logged out once when the new version lands, so push in a quiet window.
2. Confirm the health endpoint reports the new commit.
3. No database step: the new table is created by the schema on first use.
4. No new setting. OTP delivery uses the Telegram bot and WhatsApp template already live for the express transfer.
5. Tell the PMs: reading out the code is an approval; link Telegram if not yet done.

Rollback

None needed. If the OTP path misbehaves, users simply use the on-screen approval, which is unchanged. The whole two-stage batch can still be switched back with VLB_TWO_STAGE_BATCHES=0 as before; the reopen OTP works in either mode, the unlock OTP only with the two-stage batch.

8. Not included

- A destination-PM second OTP (not needed: both approvals are by the PM of the batch’s own site).
- OTP for the on-screen rejection path; a PM who wants to refuse simply does not share the code, and the code expires.
- Pulse write endpoints for transfers and new assets (separate open item, unchanged).